DISTRIBUITED SYSTEMS NOTES

Carlo Da Rome NOV. 5. 2026

CH. 1 G.P.E.

QUESTION 1: WHAT IS A DISTRIBUITED SYSTEM?

ANSWERE: A DISTRIBUITED SYSTEM IS A SET OF COMPUTERS THAT COOPERATE TO EXECUTE OPERATIONS OR TO STORE DATAS, THEY DON'T USE CLOCK FOR SINCRONIZATION, THE SINCRONIZATION IS POSSIBLE THROUGH MESSAGES.

- A LOT OF PROBLEMS IN THIS TYPE OF SYSTEM CAN BE TRACED BACK TO G.P.E. (global predicate evaluation) THAT IS EVALUATE IF THE SYSTEM SATISFIES A CONDITION.
- AN ASINCROUNOS DYSTRIBUITED SYSTEM (A.D.S.) IS A SYSTEM WITH THE FOLLOWING PROPERTIES:

1. THE M PROCESSES IN THE SYSTEM IS REPRESENTED WITH THE SET $(P_1, \ldots, P_m)$.
2. THE COMMUNICATION CHANNELS IS BASICALLY NOT FIFO, THIS MEANS THAT THE ORDER OF SENDED MESSAGES COULD BE DIFFERENT TO THE RECIEVE ORDER.
3. PROCESSES ARE STRONG CONNECTED, THIS MEANS THAT PROCESSES ARE ALL CONNECTED EACH OTHER.
4. CHANNELS ARE RELIABLE... WE CAN ASSUME THAT WE CANNOT HAVE DATA LOST OR COMPROMIZED.

5. SINCRONIZATION PASSES ONLY THROUGH MESSAGES

THIS IS A THEORETICAL MODEL, IT DOEST EXIST IN REALITY 'CAUSE IS NOT POSSIBLE TO REALIZE THAT BUT IT IS A GOOD ABSTRACTION BECAUSE.

1. IS VERY SIMPLE AND DOESN'T HAVE ANY BOUND IN TERM OF SPEED OR RETARD, IF ONE SOLUTION IS GOOD ON THIS MODEL IT IS ALSO GOOD IN REAL LIFE.

2. SOME LAYERS IN REAL SYSTEMS FOLLOW THIS MODEL (RULE 4 IS IMPOSSIBLE TO SATISFIES IN REAL LIFE)

- IN A DISTRIBUITED COMPUTATION WE CAN HAVE 3 TYPE OF EVENTS

1. INTERNAL EVENT.
   IT ONLY CHANGE THE STATE OF THE PROCESS THAT EXECUTES THE EVENT.
2. EXTERNAL EVENT.
   2.1 SEND(m)
   2.2 RECIEVE(m)

THESE TWO TYPE OF EVENTS ATTACH A MESSAGE INDICATED WITH THE LEETER "m"

CH.2 GLOBAL STATES, CUT AND CONSISTENCY

QUESTION 2: WHAT IS CONSISTENCY?

ANSWERE: IN A D.S. WE HAVE CONSISTENCY WHEN